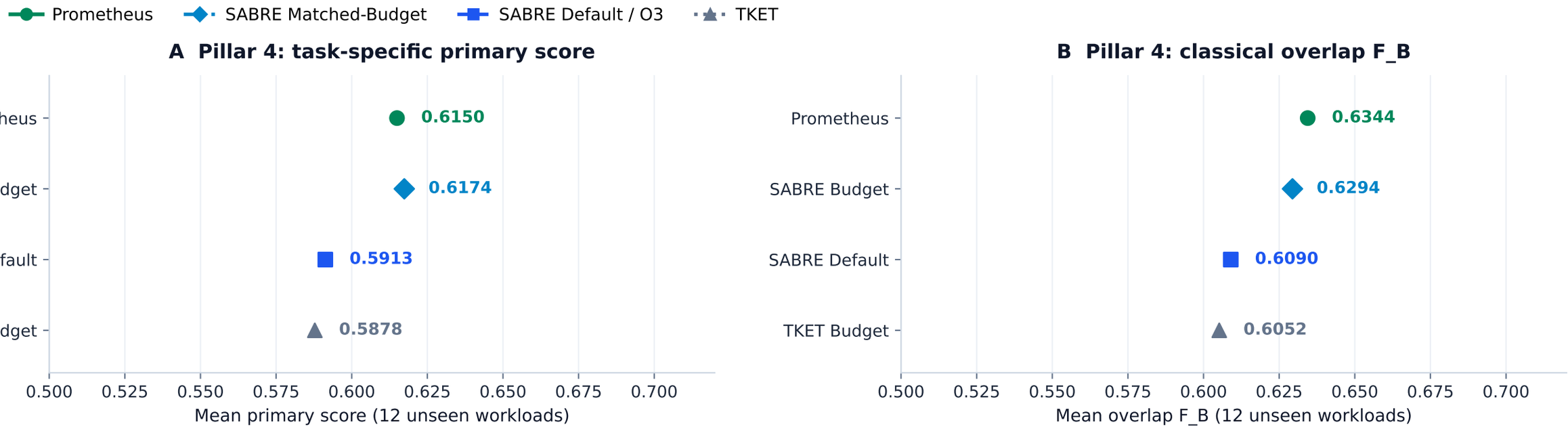
# Prometheus: equal-budget generalization and QAOA trajectories

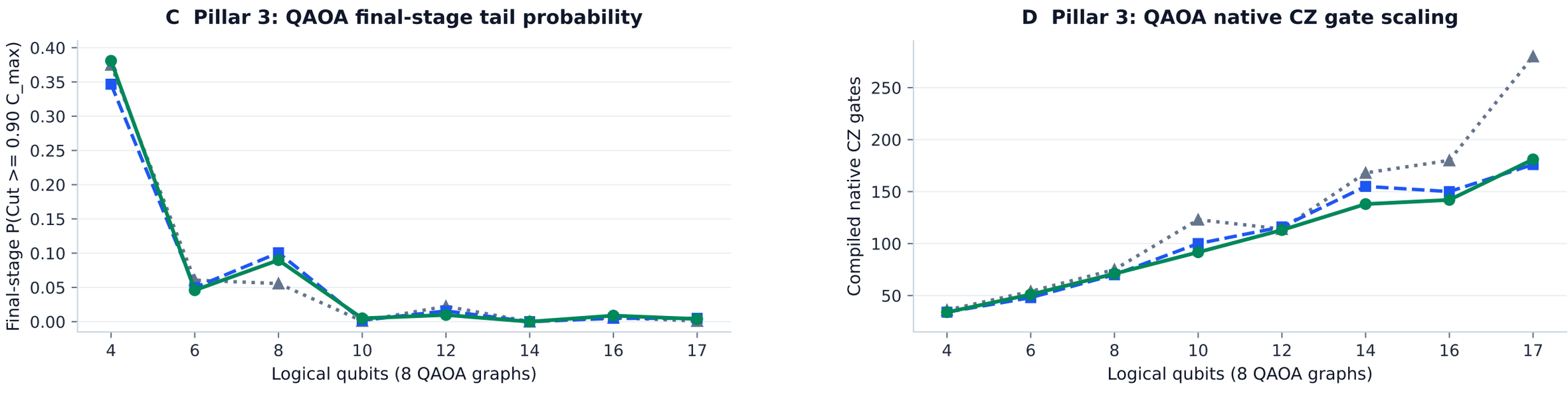
Rigetti Cepheus-1-108Q | Pillar 4: 12 unseen workloads (96 circuits) | Pillar 3: 8 QAOA graphs x 3 stages (72 circuits)

Pillar 4 matches classical CPU compile time across Block 1 (packed) + Block 2 (isolated); Pillar 3 tracks COBYLA stages.




Combines predefined family metrics across Block 1 + Block 2.

Prometheus achieves the highest mean overlap F_B (0.6344).




Final COBYLA stage mean: 0.0680 Prometheus vs 0.0656 SABRE & 0.0652 TKET.

All-24-stage mean: 102.71 Prometheus vs 106.12 SABRE (-3.22%) & 128.75 TKET (-20.23%).

**Unseen challenge gain: +4.01% vs SABRE Default (Holm p = 0.0337) | +4.62% vs TKET Budget**

Vs SABRE Matched-Budget: primary score 0.6150 vs 0.6174 (Holm p = 0.8062); classical overlap F_B 0.6344 vs 0.6294.

Pillar 3 QAOA spans 8 widths (4-17Q) x 3 stages = 24 instances (72 circuits); final-stage mean approx ratio: 0.5970 Prom, 0.6028 SABRE, 0.5991 TKET.